

## ROCK RIVER, YT, Canada

WMO: 715060

Lat: 66.9811N

Lon: 136.2181W

Elev: 2398

StdP: 13.47

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -29.2      | -25.8 | -36.7                         | 0.7 | -28.7 | -33.1 | 0.9 | -25.3 | 65.5                  | -9.4 | 59.3 | -8.6 | 4.3                   | 0    | 0.956 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.6                   | 71.7            | 56.1 | 68.2 | 54.7 | 65.2 | 53.2 | 58.2                | 68.4 | 56.3 | 65.5 | 54.5 | 62.7 | 10.1                 | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 53.8                            | 67.4 | 61.4 | 51.9 | 62.7 | 59.6 | 50.2 | 58.9 | 57.7 | 26.2          | 68.4 | 25.0 | 65.6 | 23.8 | 62.9 |                   |
|                                 |      |      |      |      |      |      |      |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 52.7              | 43.8 | 36.5 | -30.8                      | 76.6 | 4.6                | 4.4 | -34.1                                              | 79.8 | -36.8      | 82.3 | -39.4      | 84.8 | -42.7      | 87.9 |  |  |
|                   |      |      | -31.5                      | 60.4 | 4.5                | 2.3 | -34.7                                              | 62.1 | -37.3      | 63.4 | -39.8      | 64.7 | -43.1      | 66.3 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 22.3   | -4.1  | 0.6   | 2.2   | 18.1  | 37.0  | 50.9  | 54.7  | 48.2  | 36.8  | 19.5  | 4.1   | -1.4  |
|                                                                                  | DBStd       | 24.19  | 15.51 | 13.52 | 13.42 | 12.53 | 11.44 | 8.23  | 7.03  | 8.13  | 9.23  | 9.65  | 11.62 | 12.85 |
|                                                                                  | HDD50       | 10499  | 1678  | 1384  | 1482  | 957   | 423   | 88    | 37    | 131   | 403   | 945   | 1378  | 1593  |
|                                                                                  | HDD65       | 15583  | 2143  | 1804  | 1947  | 1406  | 869   | 426   | 324   | 523   | 846   | 1409  | 1828  | 2058  |
|                                                                                  | CDD50       | 398    | 0     | 0     | 0     | 0     | 19    | 114   | 182   | 75    | 7     | 1     | 0     | 0     |
|                                                                                  | CDD65       | 10     | 0     | 0     | 0     | 0     | 0     | 2     | 5     | 3     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 47     | 0     | 0     | 0     | 0     | 0     | 11    | 20    | 16    | 0     | 0     | 0     | 0     |
|                                                                                  | CDH80       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 10.7   | 9.2   | 7.9   | 8.9   | 11.8  | 13.8  | 13.5  | 10.6  | 11.2  | 10.8  | 11.4  | 9.6   | 9.7   |
| Precipitation                                                                    | PrecAvg     | 14.5   | 0.7   | 0.6   | 0.6   | 0.5   | 0.8   | 1.8   | 2.6   | 2.4   | 1.7   | 1.2   | 0.8   | 0.8   |
|                                                                                  | PrecMax     | 17.0   | 1.7   | 1.7   | 1.4   | 1.0   | 1.6   | 3.4   | 4.4   | 4.8   | 2.6   | 2.0   | 2.0   | 2.0   |
|                                                                                  | PrecMin     | 11.9   | 0.2   | 0.1   | 0.2   | 0.1   | 0.2   | 0.4   | 1.1   | 0.8   | 1.1   | 0.2   | 0.1   | 0.3   |
|                                                                                  | PrecStd     | 1.3    | 0.4   | 0.4   | 0.3   | 0.2   | 0.4   | 0.7   | 0.7   | 0.9   | 0.4   | 0.4   | 0.4   | 0.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 36.7  | 35.1  | 36.2  | 49.6  | 67.2  | 75.3  | 77.0  | 75.6  | 63.1  | 46.9  | 35.2  | 34.4  |
|                                                                                  |             | MCWB   | 32.2  | 29.5  | 30.8  | 40.0  | 50.1  | 55.4  | 58.7  | 60.0  | 48.5  | 40.6  | 30.1  | 29.7  |
|                                                                                  | 2%          | DB     | 29.3  | 29.3  | 30.1  | 43.7  | 62.4  | 70.1  | 72.3  | 69.1  | 56.9  | 40.1  | 28.8  | 27.7  |
|                                                                                  |             | MCWB   | 25.9  | 25.2  | 26.3  | 36.2  | 48.1  | 53.5  | 57.0  | 57.0  | 47.0  | 35.4  | 25.7  | 24.8  |
|                                                                                  | 5%          | DB     | 24.3  | 24.2  | 25.9  | 39.6  | 57.9  | 66.8  | 68.6  | 64.2  | 53.1  | 36.0  | 25.1  | 23.0  |
|                                                                                  |             | MCWB   | 21.4  | 21.3  | 22.5  | 33.5  | 45.8  | 52.7  | 55.7  | 54.4  | 45.2  | 32.6  | 22.9  | 20.7  |
|                                                                                  | 10%         | DB     | 19.4  | 20.0  | 21.4  | 35.6  | 53.6  | 63.4  | 65.7  | 60.1  | 49.5  | 32.5  | 20.7  | 17.8  |
|                                                                                  |             | MCWB   | 17.5  | 17.8  | 18.7  | 30.8  | 43.8  | 51.2  | 54.3  | 52.2  | 43.3  | 30.0  | 19.1  | 16.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 32.6  | 30.1  | 31.7  | 40.1  | 51.2  | 59.0  | 60.3  | 61.7  | 50.8  | 41.2  | 30.5  | 30.5  |
|                                                                                  |             | MCDB   | 36.4  | 35.4  | 35.4  | 48.8  | 64.9  | 69.9  | 72.0  | 71.5  | 59.6  | 46.8  | 34.6  | 33.2  |
|                                                                                  | 2%          | WB     | 25.9  | 25.4  | 26.7  | 36.8  | 48.7  | 55.7  | 58.5  | 57.8  | 47.9  | 35.9  | 26.1  | 24.8  |
|                                                                                  |             | MCDB   | 28.8  | 28.8  | 30.1  | 43.3  | 60.5  | 66.2  | 69.0  | 67.2  | 54.6  | 39.3  | 28.4  | 27.2  |
|                                                                                  | 5%          | WB     | 21.8  | 21.5  | 22.7  | 33.7  | 46.5  | 53.7  | 57.0  | 55.2  | 45.9  | 32.9  | 22.8  | 20.6  |
|                                                                                  |             | MCDB   | 24.2  | 23.9  | 25.5  | 39.2  | 56.9  | 64.3  | 66.6  | 63.0  | 51.8  | 35.8  | 25.2  | 22.9  |
|                                                                                  | 10%         | WB     | 17.4  | 17.9  | 18.8  | 30.8  | 44.2  | 52.0  | 55.4  | 52.8  | 44.0  | 30.3  | 19.1  | 16.4  |
|                                                                                  |             | MCDB   | 19.6  | 19.9  | 21.4  | 35.3  | 53.0  | 61.8  | 63.6  | 59.3  | 48.8  | 32.1  | 20.6  | 17.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.6  | 12.2  | 12.3  | 12.7  | 13.6  | 14.7  | 13.6  | 12.1  | 10.4  | 8.2   | 10.0  | 11.2  |
|                                                                                  |             | MCDBR  | 14.4  | 16.3  | 16.7  | 15.2  | 17.8  | 18.1  | 17.7  | 17.2  | 15.8  | 11.6  | 13.6  | 16.7  |
|                                                                                  |             | MCWBR  | 12.1  | 13.5  | 13.9  | 10.2  | 9.4   | 8.5   | 8.4   | 8.9   | 8.8   | 8.2   | 11.3  | 14.2  |
|                                                                                  | 5% WB       | MCDBR  | 14.5  | 16.0  | 16.3  | 15.1  | 16.8  | 16.7  | 16.4  | 16.0  | 13.6  | 11.0  | 13.4  | 15.8  |
|                                                                                  |             | MCWBR  | 12.5  | 13.6  | 13.9  | 10.2  | 9.2   | 8.6   | 8.6   | 9.0   | 8.1   | 8.0   | 11.4  | 13.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.159  | 0.218 | 0.235 | 0.286 | 0.322 | 0.330 | 0.349 | 0.322 | 0.280 | 0.222 | 0.147 | N/A   |       |
|                                                                                  | taud        | 1.906  | 2.159 | 2.273 | 2.192 | 2.275 | 2.420 | 2.381 | 2.494 | 2.538 | 2.290 | 1.935 | N/A   |       |
|                                                                                  | Ebn at Noon | 73     | 207   | 269   | 277   | 277   | 278   | 267   | 265   | 254   | 206   | 76    | 0     |       |
|                                                                                  | Edh at Noon | 4      | 15    | 23    | 33    | 35    | 31    | 31    | 25    | 19    | 13    | 3     | 0     |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 23     | 217   | 781   | 1517  | 1800  | 1848  | 1488  | 1087  | 637   | 248   | 47    | 0     |       |
|                                                                                  | RadStd      | 2      | 13    | 37    | 57    | 156   | 167   | 117   | 103   | 60    | 27    | 4     | 0     |       |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)  | (b)                    | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only           |       | +1.24   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (3 neighbors) |       | +2.03   | +3.51  | +3.69   | N/A   | N/A   | N/A         | -821  | -839  | N/A   |

Nomenclature: See separate page